
ELK336-Quantum Computation

MIDTERM-TAKE HOME

Due 09.05.2023

- The students are obligated to do the homework themselves.
 - The copied homework will not be considered or graded.
 - Homework should be ready before 23.00 on the ninova in electronic form.
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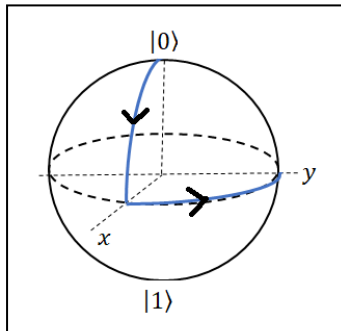
1- The spectral theorem asserts that any normal operator has a diagonal representation with respect to its basis states. Therefore, a normal operator A can be written as

$A = \sum_i \lambda_i |u_i\rangle\langle u_i|$ where λ_i are the eigenvalues and $|u_i\rangle$ are the corresponding eigen basis.

- (a) Find the eigen basis of Pauli- X operator and write in the $X = \sum_i \lambda_i |u_i\rangle\langle u_i|$ form. (15p)
(b) Find the matrix representation of the square root $\sqrt{X}$ of the Pauli- X operator. (10p)

2- (a) Show that $HZH = X$ and $HYH = -Y$ by matrix representation. (10p.)

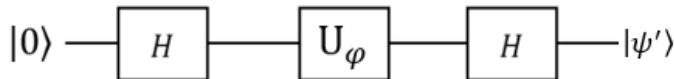
- (b) Show that $R_v = e^{-iH_v t}$ unitary rotation equals $R_v = \cos \frac{\theta}{2} \mathbb{1} - i \sin \frac{\theta}{2} \sigma_v$ where H_v is the rotation Hamiltonian with, $H_v = \frac{\omega}{2} \sigma_v$ by using $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$ expansion.
(Here, σ_v are the Pauli operators; $v; x, y, z$ and take $\omega t = \theta$). (15 p.)



3- A quantum state initialized in $|0\rangle$, follows the path in the figure (left panel) and reaches $|0\rangle \rightarrow |\psi'\rangle$.

- (a) Which unitary rotations performed for the path? Construct the matrix representations of the respective rotations. (15p.)
(b) Draw the circuit diagram gives $|\psi'\rangle$ and calculate $|\psi'\rangle$. (10 p.)

4- Given the following quantum circuit ;



- (a) Find the $|\psi'\rangle$ output state. Here $U_\varphi = |0\rangle\langle 0| + e^{i\varphi}|1\rangle\langle 1|$ is a relative phase gate. (10 p.)
(b) Show that the relative phase φ applied by U_φ can be calculated as $\varphi = \cos^{-1}[p_0 - p_1]$ where p_0 and p_1 are the probabilities finding the $|\psi'\rangle$ state in $|0\rangle$ and $|1\rangle$ respectively. (15 p.)